

The Firm Next Door: Using Satellite Images to Study Local Information Advantage

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ABSTRACT

We use novel satellite data that track the number of cars in the parking lots of 92,668 stores for 71 publicly listed U.S. retailers to study the local information advantage of institutional investors. We establish car counts as a timely measure of store-level performance and find that institutional investors adjust their holdings in response to the performance of local stores, and that these trades are profitable on average. These results suggest that local investors have an advantage when processing information about nearby operations. However, some institutional investors do not adjust for the *quality* of their

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local information and continue to rely on local signals even when they are poor predictors of firm performance and returns. This overreliance on poor local information is reduced for institutional investors with greater industry expertise and those with greater incentives to maximize short-term trading profits.

JEL codes: G23, G30, G40, G41, K00, L81, M40, M41, M48, O33

Keywords: satellite images; store-level performance; institutional investors; local advantage; overweighting; processing costs; alternative data; big data; emerging technologies

1. Introduction

A large stream of literature has concluded that local investors benefit from an information advantage (e.g., Coval and Moskowitz [2001], Bernile, Kumar, and Sulaeman [2015]). However, these studies measure local information advantage indirectly, using the profitability of local trades, making it difficult to identify the information to which local investors have access. We build on these prior studies by using novel satellite image data to measure intra-firm variation in the performance signals of stores located in different geographic areas. As a result, we can pinpoint the information signals that institutional investors incorporate into their trades, and our results suggest that one source of local information advantage is information about the performance of nearby stores. We also study the extent to which institutional investors appropriately integrate this local information into their trades.

Prior research has been largely agnostic as to the type of information that drives local information advantage. On one hand, local exposure to a firm may provide institutional investors with information about *firm-wide* performance and events. For example, local investors may gain private company-wide information through greater access to firm executives. Additionally, exposure to local stores might increase investors' knowledge about the firm's policies and business model, which have implications for firm performance. On the other hand, geographic proximity may lead local investors to have an information advantage relating solely to *local* operations and performance. Our data and empirical tests explore this latter type of local information.

In the framework developed in Blankespoor, Dehaan, and Marinovic [2020, 2019], an information signal effectively becomes private information if the costs to become aware of, acquire, or integrate (analyze) that signal become too high relative to the expected benefits of using it. In our setting, geographic proximity, social ties, or institutional knowledge may lower the costs that institutional investors face to obtain information about local stores, leading them to have systematically more information about local operations relative to nonlocal investors. For example, institutional investors may observe store busyness when they shop. They may know local

employees or managers who mention store performance, or have ties with reporters who can keep them apprised of local updates. Or they may be invited to events, engage in policy making, or participate in social clubs and organizations that provide information about factors that cause shifts in supply and demand, for example, the closure or opening of a competitor, or new state or city ordinances. On the other hand, nonlocal institutional investors wishing to obtain the same information would need to purchase business intelligence, pore over public reports, or travel to the region, and even then, might not observe signals that local investors can.¹

To investigate whether institutional investors have an information advantage relating to local operations, we use satellite image data on the number of cars in nearby store parking lots as a measure of store-level performance. We validate the data by showing that changes in store-level car counts are positively associated with quarterly changes in firm-level sales and net income, and with abnormal returns. Although we do not expect local institutional investors to directly observe car counts, we expect the local information signals they *do* observe to be positively correlated with local car counts through their joint relation with the performance of local stores. Further, we ensure that none of the investors that we study respond directly to the satellite data itself by restricting our sample period to end before the data became commercially available. Our main sample includes 92,668 stores of 71 publicly listed U.S. retailers during the years 2009–2015.²

We define an institutional investor as local to a store if they are located within 50 km.³ If investors have greater information about local operations, they should respond more to the performance of local stores than to that of nonlocal stores. We provide evidence of this intuition by showing that investors adjust their stock holdings in line with the car counts of local stores, but not with the concurrent performance of the overall firm. On average, a one-standard-deviation increase in local car counts is related to investors increasing their holdings by 9–14%. Further, we provide evidence that trades that incorporate local car count information yield annualized abnormal returns of roughly 2%. These abnormal returns are comparable in economic magnitude to those documented in the previous literature (Coval and Moskowitz [2001], Baik, Kang, and Kim [2010]) and suggest

¹ In principle, these same factors should also apply to retail investors, even though they are likely less sophisticated and have fewer resources at hand. We test our hypotheses using institutional investors because we have access to location and trading data for only these investors.

² When Orbital Insight was formed in 2013, it developed its algorithms by analyzing past satellite images. Thus, when the Orbital Insight data first became available for purchase in 2015, the company already had historical data starting in 2009. We use only the 2009–2015 data so that, by construction, none of the investors in our main tests were responding directly to the satellite data.

³ In the online appendix, section A, we show that our results are robust to using alternative distances to define local investors including: 100 km, 100 miles, within the same state, and within the same “super county,” where a super county includes all counties with overlapping zip codes.

that information about local operations is one source of local information advantage.

We investigate alternative explanations for our results. We show that the response to local store information is incremental to the response to firm-level performance, ruling out the possibility that our results are driven by investors responding to concurrent firm-level signals. We also find that institutional investors respond to local car counts incremental to local economic conditions, suggesting local information is store-specific. Additional analyses show no evidence of reverse causality whereby investors' trades drive local performance, for example through heightened monitoring, or that local information advantage is purely driven by local media.

We conduct a series of cross-sectional tests. First, we find that the response to local information is greater in regions with stronger social ties, consistent with these ties providing access to local performance information. Second, we find that institutional investors who are industry specialists (Kacperczyk, Sialm, and Zheng [2005]) are more responsive to local information, consistent with their industry expertise making it easier for them to interpret and trade on local signals (Blankespoor, Dehaan, and Marinovic [2020]). Further, we find that institutional investors who charge their clients performance fees respond more to local information, consistent with them having greater incentives to use local information when they stand to benefit personally. Finally, we find that the response to local information is largest for nonindexer institutional investors (Bushee [2001]), whose active investment strategies are more reliant on timely information signals.

We also examine whether investors appropriately weight their local signals. To do so, we identify local car counts that are *ex ante* poor signals of firm performance. A rational investor with no constraints should place a lower weight on local signals of predictably poor quality. However, the processing costs necessary to determine the quality and optimal weights of local signals may exceed the trading benefits of doing so (Blankespoor, Dehaan, and Marinovic [2020, 2019]). Further, behavioral biases, such as the representativeness heuristic (Tversky and Kahneman [1974]) and the law of small numbers (Tversky and Kahneman [1971]), may lead investors to overgeneralize their local signals. Consistent with both of these explanations, our results indicate that, on average, the institutional investors in our sample do not fully incorporate information on the quality of their local signals when weighting them in their trading decisions. As a result, they continue to trade on local signals even when they are no longer significant predictors of future performance or returns. However, we find evidence that institutional investors who charge performance fees and those who specialize in the retail industry are less likely to trade on poor-quality local information. This is consistent with these investors having particularly high incentives to use local information to trade profitably and facing lower costs to evaluate the quality of their local signals, respectively.

Finally, we find that after the satellite data used in this study became commercially available, institutional investors who had access to the data began incorporating the car count information of *all* stores into their trades, with no incremental response to the performance of local stores. These results suggest that access to big data enables investors to overcome their nonlocal information disadvantage, providing additional evidence on the benefits of big data (Zhu [2019]). As timely and informative data sources like the one used in this study become more widely available, the information playing field between local and nonlocal investors may become more level. However, we caution that this last result applies only to institutional investors with the means to purchase the Orbital Insight data, while retail investors and smaller institutional investors may be falling farther behind in the informational arms race (Katona et al. [2019]).

Our results are important for several reasons. First, while prior studies have provided evidence consistent with a generic local information advantage (Coval and Moskowitz [2001], Baik, Kang, and Kim [2010], Bernile, Kumar, and Sulaeman [2015]), we pinpoint one specific type of information that contributes to this advantage: information about the performance of local stores. Our unique design uses a measure of performance that varies within firm and a definition of localness that varies within an investor–firm pair to show that local investors have a particular advantage when acquiring information about nearby operations. Thus, we provide a more direct test of local information advantage than studies that focus solely on the profitability of local trades.

Second, by identifying one type of relevant local information, we can study how institutional investors incorporate this information into their investment decisions. Our evidence suggests that some institutional investors rely too heavily on poor-quality local information signals and continue to trade on them even when they would be better off ignoring them. These results indicate that although local institutional investors may have an information advantage on average, they still face barriers associated with integrating their local information. Our cross-sectional tests on the types of investors who use poor-quality local information provide some evidence that this may be due to the costs of evaluating local signal quality, because investors with greater incentives to trade profitably or lower costs to evaluate information are more discerning in the local information that they respond to (Blankespoor et al. [2019]). However, we cannot rule out the possibility that behavioral biases also play a part.

Finally, our study adds to the emerging field that uses big data to examine economic questions generally (Froot et al. [2017]), as well as research examining the impact of big data in particular (Zhu [2019]). Our study is especially timely given that investors now routinely supplement the formal financial statements with less traditional information sources like the data used in this study (Goldman Sachs [2016]).

2. *Related Literature and Hypotheses*

A number of studies have found evidence of local information advantage for several stakeholders, including institutional investors (Coval and Moskowitz [2001], Bae, Stulz, and Tan [2008], Baik, Kang, and Kim [2010], Bernile, Kumar, and Sulaeman [2015], Chi and Shanthikumar [2016]); however, these studies are largely agnostic as to the source of this information. We predict that one source of local information advantage is greater information about the performance of local stores, for example as a result of geographic proximity, social ties, or institutional knowledge.

Blankespoor, Dehaan, and Marinovic [2020, 2019] describe the costs that investors incur to process information signals as a three-step process: awareness costs of monitoring for the existence of a signal; acquisition costs of obtaining the signal; and integration costs of analyzing the signal's implications for firm value.⁴ If local institutional investors face lower costs during any of these stages, they will have an information advantage. For example, local institutional investors may observe the busyness of local stores when they do their shopping. Their knowledge of local economic conditions and institutions, such as local competitors or ordinances, may help them predict and interpret information about local performance. They may acquire information about store performance and hiring through social ties with managers and other employees. Additionally, local media outlets or analysts may disproportionately gather and disseminate information about local business activities (Miller and Shanthikumar [2015]) and local investors may consume more of these local news sources or even have ties with local reporters. Further, institutional investors may have especially valuable local information because their financial expertise and higher than average education level may lead them to have greater ties with social elites who can provide more precise information on store-level performance or institutional changes, including business leaders and policy makers, through venues such as country clubs or leadership committees. In addition, institutional investors are sophisticated users of financial information, meaning they may face lower costs to integrate local information than noninstitutional investors, especially because they are likely to pay more attention to firms in which they are actively investing. In short, we expect institutional investors to benefit from a significant local information advantage, providing us with a powerful empirical setting.⁵

⁴ Investors who can never access a particular signal have infinitely high awareness or acquisition costs.

⁵ In contrast, Miller and Shanthikumar [2015] argue that because institutional investors have many other information sources at their disposal, local information may constitute a smaller proportion of their information set. If the proportion, rather than the amount, of local information is the driving factor in whether institutional investors use local information, then this should decrease our ability to find significant responses to local information in our setting.

The prior literature has explored some aspects of local information advantage, in particular for institutional investors, but has not identified the information that these local investors possess. Ellis, Madureira, and Underwood [2019] find that mutual funds' investments are more profitable when travel time between the fund and firm headquarters decreases. Their results may capture increases in the ease of observing local operations or may be driven by firm-level information obtained from executives (Solomon and Soltes [2015]). Engelberg and Parsons [2011] and Miller and Shanthikumar [2015] find that investors trade profitably in firms covered by local newspapers, suggesting that newspapers may be a valuable source of information to investors. However, it is not clear whether this information is firm-wide or purely local. Finally, Bernile, Kumar, and Sulaeman [2015] show that firms with similar regional exposure have correlated performance and institutional investors' holdings exhibit superior stock performance when they have high local exposure, consistent with local information advantage. However, Bernile, Kumar, and Sulaeman [2015] do not directly measure local information and therefore cannot distinguish whether institutional investors have greater access to information about local operations, or whether local exposure provides investors access to firm-level information.

We add to this literature by testing whether institutional investors have greater information about the performance of local stores. We use the number of cars in store parking lots as a proxy for store performance and define an investor as *Local* to a store if they are located nearby (i.e., within a range of 50 km). An investor may be local to some stores within a firm and nonlocal to others. Figure 1 provides a graphical depiction of this definition. If institutional investors have more information about local stores, then their stock holdings should be incrementally responsive to local store performance after controlling for overall firm performance.

H1: Changes in institutional investors' stock holdings have a positive relation with changes in the car counts of local stores incremental to car counts of the firm overall.

Although we predict that local investors will have an information advantage relating to the performance of local stores on average, we expect there to be variation in the extent to which certain investors have access to, interpret, and choose to use their local information. First, there is likely to be variation across investors in the extent to which they have access to local information. Although all local investors are geographically proximate to local operations, some may be located in areas in which there is less social interaction and engagement. Because social ties with other local residents are likely to be a valuable source of local performance information, such investors may be less able to obtain local information. As a result, we expect that investors located in areas with strong social ties will be most likely to obtain, and thus respond to, local performance information.

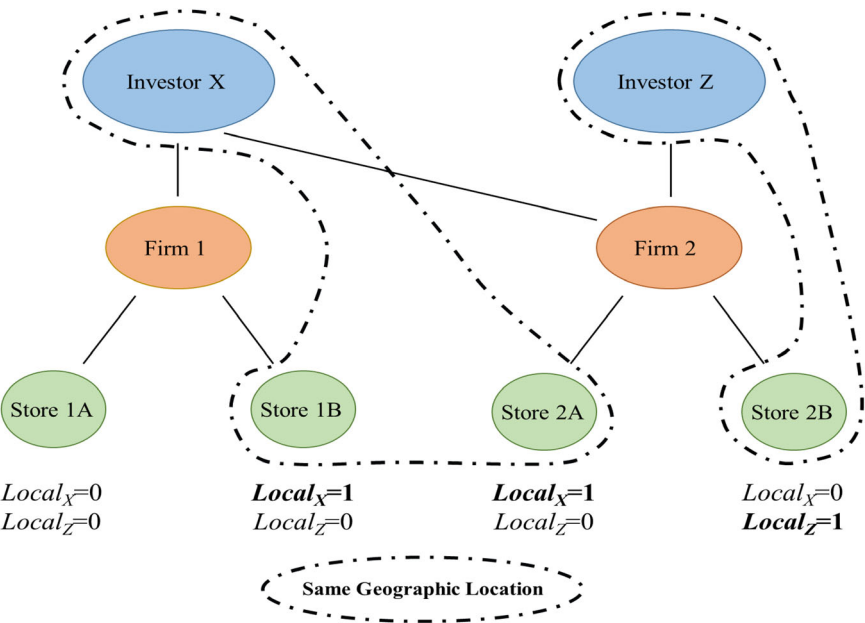


FIG 1.—Data structure. This figure plots the structure of the data used in the paper. Investors are local to individual stores within a firm if they are located within the same geographic location (within 50 km). The car counts of only local stores (for a given investor–firm–quarter pair) are used to define *Car_Chg_Local*. The car counts of all stores within a firm–quarter are used to define *Car_Chg_Firm*.

H2a: Institutional investors are more responsive to local performance information when they are located in communities with stronger social ties.

Next, we expect there to be variation in the extent to which institutional investors can interpret local information and therefore integrate it into their trading decisions (Blankespoor et al. [2019]). In particular, we expect that industry expertise may allow institutional investors to analyze local information about firms in their industry of expertise at a lower cost (Kacperczyk, Sialm, and Zheng [2005]). Such investors would therefore be more likely to incorporate local performance information about these firms into their trades.

H2b: Institutional investors are more responsive to local performance information when they have greater industry expertise.

Finally, we expect that there is likely to be variation across institutional investors in the extent to which they choose to acquire and use their local information. Because acquiring and interpreting local information entails costs, investors must weigh the incremental net benefit of local information. As a result, institutional investors will have greater incentives to extract

and use available local information if they stand to gain more personally from maximizing their trading decisions. For example, institutional investors who charge performance fees to their clients receive compensation based on the capital gains of their clients' assets and therefore benefit more in the short term from profitable trading decisions. Conversely, the immediate payoffs of those who do not charge performance fees are significantly less sensitive to how their investments perform. As a result, we expect institutional investors who charge performance fees to have greater incentives to identify and trade on valuable local information.

H2c: Institutional investors are more responsive to local performance information when they charge performance fees.

Similarly, institutional investors are unlikely to use their local information if they do not rely upon information-based trading strategies. Institutional investors who rely on passive investing strategies, in particular index investors, purchase and hold investments for the long run, and do not respond to firm-level information in the short run. Thus, these types of investors will be less likely to rely on local information.

H2d: Institutional investors are more responsive to local performance information when they rely on information-based (nonindexing) trading strategies.

In addition, although we expect variation in the extent to which investors incorporate local information into their trades, we expect that the local information that they *do* trade on will be beneficial. Specifically, we predict that the local performance information that we measure contributes to the local information advantage documented in prior research (Coval and Moskowitz [2001]) and that trades based on this information are profitable. Conversely, local information might be unprofitable if investors overweight it (Cheng and Eshleman [2014]), if the incremental information is negligible, or if local information is costly to analyze (Blankespoor et al. [2019]). We predict:

H3: Trades based on local store information are profitable.

Finally, investors may not always appropriately weight their local signals. A rational investor with no constraints should rely on local signals less when they are of predictably poor quality (Griffin and Tversky [1992]). However, local investors sometimes make unprofitable trades for noninformation-driven reasons (Hochberg and Rauh [2013]), and even sophisticated market participants may weight information signals inefficiently (Chen and Jiang [2006]). Blankespoor et al. [2019] propose that rational investors face costs to analyze information, and they document the existence of such integration costs in a sample of retail investors. In our setting, it may be costly for institutional investors to assess the quality of their local signals. If these costs are high enough relative to the expected benefits of using local signals, then institutional investors may respond equally to all local

signals, regardless of their quality. For example, H2 predicted that institutional investors without industry expertise might face higher costs to evaluate and interpret local information and that institutional investors who do not charge performance fees face lower incentives to profitably trade on their local information. As a result, a low net benefit of determining local signal quality might prevent either group from distinguishing between high- and low-quality local signals.⁶ Further, prior studies have documented behavioral biases, such as the representativeness heuristic (Tversky and Kahneman [1974]) and the law of small numbers (Tversky and Kahneman [1971]), which may lead investors to overgeneralize their local signals and assume that they are of higher quality than they really are. As a result of either of these mechanisms—processing costs or behavioral biases—institutional investors may rely too heavily on poor-quality local signals when making inferences about firm-level performance. We therefore identify cases when local signals are *ex ante* expected to be of poor quality and examine whether institutional investors rely too heavily on these poor-quality signals. However, because these processing costs or behavioral biases may be absent on average, we state H4 in null form:

H4: Institutional investors properly discount local information in their trading decisions when it is of poor quality.

3. *Empirical Setting and Variable Definitions*

3.1 SATELLITE DATA

We use a unique data set containing store-level car counts extracted from satellite images by Orbital Insight, a geospatial analytics company whose clients include asset management firms, U.S. government agencies, and nonprofits.⁷ The satellite images used are roughly 50 cm² per pixel and high resolution enough to distinguish cars from neighboring objects. Orbital Insight uses a four-step process to extract car count information from these images. First, Orbital Insight identifies the coordinates of every store for each tracked retailer. These coordinates are updated periodically to add new stores and remove terminated stores. Second, for each store, Orbital Insight identifies the relevant area of the parking lot where the machine vision algorithm should count cars. Each store is assigned a contamination score based on the likelihood that the algorithm incorrectly assigns cars to

⁶ H2 also predicted that nonindexer investors would be less likely to use local information and that institutional investors in areas with weak social ties would be less likely to obtain local information. However, there is no reason to expect either of these attributes to be tied to differences in how investors respond to high- and low-quality local signals once they have obtained them. In the former case, this is because indexer investors do not use local information at all. In the latter, it is because there is no reason to believe the strength of investors' social ties will be tied to their ability or incentive to determine local signal quality.

⁷ We use Orbital Insight Consumer Insight Version 4.2.1.

stores, and stores with covered parking lots are excluded.⁸ Third, Orbital Insight adjusts their car count data based on time of day to ensure that images are comparable and excludes images taken outside of store hours. Finally, they perform cleaning procedures, such as removing the portions of images blocked by clouds. Each night, a proprietary algorithm based on convolutional neural network machine vision analyzes images collected during the day, and the final car count data are provided to clients the following morning.⁹

In table 1 and figure 2(a), we provide descriptive information for the satellite data that span from March 2009 to June 2015 when the satellite data became available for purchase. We conduct our main analyses using the period before commercial availability to ensure we do not capture investors' responses to the data itself.¹⁰ After merging with accounting variables from Compustat and 13F holdings, the sample covers 92,668 unique stores distributed across 49 states, Puerto Rico, and Washington, DC. As can be seen in figure 2(a), these stores are dispersed widely across the United States, with greater density in the eastern United States, consistent with population patterns. The 92,668 stores are owned by 71 publicly listed retailers and have 654,874 quarterly observations (an average of seven quarters per store). The largest retailer has 10,706 stores while the smallest has 51. However, even the smallest retailers have a relatively wide geographic spread, with the least geographically dispersed retailer operating in 10 states, and the most dispersed retailer operating in 49 states and Washington, DC. As seen in panel D of table 1, the retailers in our sample are larger than other firms (retailers) covered by Compustat, with average assets of \$2.1 billion compared to average assets of \$1.2 billion (\$1.2 billion) for all other Compustat firms (retailers) in the same time period.

3.2 EMPIRICAL DEFINITION OF LOCAL INVESTORS AND PROXY FOR LOCAL PERFORMANCE INFORMATION

We measure whether investors are local at a more granular level than prior studies by identifying whether an investor is local to individual stores, which allows an investor to be both local and nonlocal to different stores within the same firm. An investor is local to a store (*Local*) if they are located within 50 km of that store. Similar to prior studies, such as Coval and Moskowitz [2001], this definition relies on the intuition that geographically

⁸ In untabulated tests, we exclude high contamination stores and our results remain similar.

⁹ Although individual stores are not observed every day, Orbital Insight uses current store-level car counts and historical patterns to estimate a daily firm-level car count measure. Our study uses individual store-level data.

¹⁰ Katona et al. [2019] provide evidence that investors respond to information from satellite data from another data vendor, RS Metrics, and these data were available for purchase as early as 2011. In untabulated robustness tests, we identify firms in our sample with RS Metrics coverage according to table 1 of Katona et al. [2019] and exclude firm quarters for which RS Metrics data were commercially available. Inferences are not sensitive to this exclusion.

TABLE 1
Satellite Image Data

Panel A: Sample selection				Store-Quarter Observations	Unique Firms	Unique Stores	
(1) Quarterly average car counts from Orbital Insight				1,284,151	80	106,800	
(2) Eliminate observations missing accounting data from Compustat, analyst data from IBES, and price data from CRSP				1,163,341	78	104,715	
(3) After merging with 13F data				1,116,072	71	101,471	
(4) Retain period before June 2015 when the data are made commercially available				654,874	71	92,668	
(5) Aggregate store car counts to the firm level and eliminate firm-quarters missing four-quarter lagged car count changes				1,218	71	N/A	
Panel B: Store-quarter observations per geographic location							
State	Total	Unique Firms	Unique Stores	State	Total	Unique Firms	Unique Stores
AK	1,044	21	96	NC	23,757	66	3,784
AL	8,310	60	1,932	ND	1,248	40	193
AR	5,112	60	983	NE	4,468	53	598
AZ	29,736	63	2,186	NH	1,820	45	454
CA	100,745	59	8,386	NJ	13,403	55	1,980
CO	22,680	64	1,956	NM	6,563	52	729
CT	5,272	53	990	NV	13,554	59	919
DC	581	10	34	NY	19,676	59	3,276
DE	1,161	49	280	OH	14,429	66	3,867
FL	60,317	64	6,621	OK	9,583	63	1,260
GA	20,307	67	3,695	OR	7,660	52	992
IA	4,735	56	993	PA	16,481	62	3,536
ID	4,211	53	433	PR	8	1	1
IL	18,087	66	3,496	RI	2,293	47	300
IN	11,852	66	2,336	SC	13,239	61	1,989
KS	6,610	64	1,131	SD	1,153	46	233
KY	5,785	63	1,571	TN	13,991	65	2,445
LA	11,272	63	1,794	TX	62,932	68	8,540
MA	13,695	55	1,642	UT	7,847	56	844
MD	9,279	59	1,517	VA	19,517	65	2,930
ME	1,683	47	441	VT	552	33	146
MI	11,269	60	3,003	WA	10,059	52	1,681
MN	9,893	57	1,199	WI	6,353	59	1,216
MO	13,456	64	1,967	WV	1,119	46	541
MS	4,404	54	1,093	WY	638	34	19
MT	1,035	41	247				

(Continued)

TABLE 1—*Continued*

Panel C: Store-quarter observations over time

Year	Total	Unique Firms	Unique Stores
2009	30,904	52	21,007
2010	99,760	61	47,598
2011	103,440	62	50,896
2012	116,913	63	54,396
2013	110,710	68	51,272
2014	107,025	70	55,305
2015	86,122	71	52,441
Total	654,874	71	92,668

Panel D: Sample retailers versus other firms

Characteristics (Mean)	Sample Retailers	Nonsample Retailers	All Other Compustat
Assets (\$millions)	2,128.0	1,249.1	1,237.9
Market Cap (\$millions)	2,546.8	1,200	985.0
MTB	2.8	2.5	2.1
Liabilities/Assets	0.5	0.6	0.5
Revenue (\$millions)	19,429.0	5,281.6	2,856.1
NI (\$millions)	469.4	224.3	162.2
Loss	0.1	0.3	0.5

This table provides information about the sample for the Orbital Insight satellite data used in the paper. Panel A presents the sample selection process. Sample firms must have the following data available from Compustat: sales, net income, SG&A expense, stockholders' equity, and total assets (Step 2). Satellite data are only used from the period before they became commercially available (Step 4), so that our results do not capture investors' response to the data itself. Panel B (C) summarizes store-quarter observations for each state (year). Panel D compares characteristics of retailers in our sample with those of other firms and other retailers in Compustat, where other retailers are identified as firms with the same two-digit SIC codes as those in our sample: 25, 28, 51–59, and 75.

close locations are more physically accessible and are likely to have joint institutional and economic environments and social networks. Analyses reported in the online appendix, section A, show our results are not sensitive to the local definition that we choose.¹¹ About 2% of investor–store-quarter observations in our sample are local using this definition.

Next, we use car count information as an empirical proxy for the performance of individual stores. Car counts are likely to be correlated with foot traffic, which is positively associated with retail sales and store performance (Perdikaki, Kesavan, and Swaminathan [2012]). During our main sample period (prior to the satellite data becoming available), it is unlikely that many investors would systematically track cars in parking lots. However, local institutional investors may observe a variety of signals that are correlated with store-level performance. They may observe store busyness when they shop, learn about store activities through social ties with store

¹¹ The online appendix, table A1, shows our results are consistent when instead using 100 km, 100 miles, no distance restriction within the same state, or when the investor is located within the same “super county” as the store, where a super county combines all counties with overlapping zip codes.

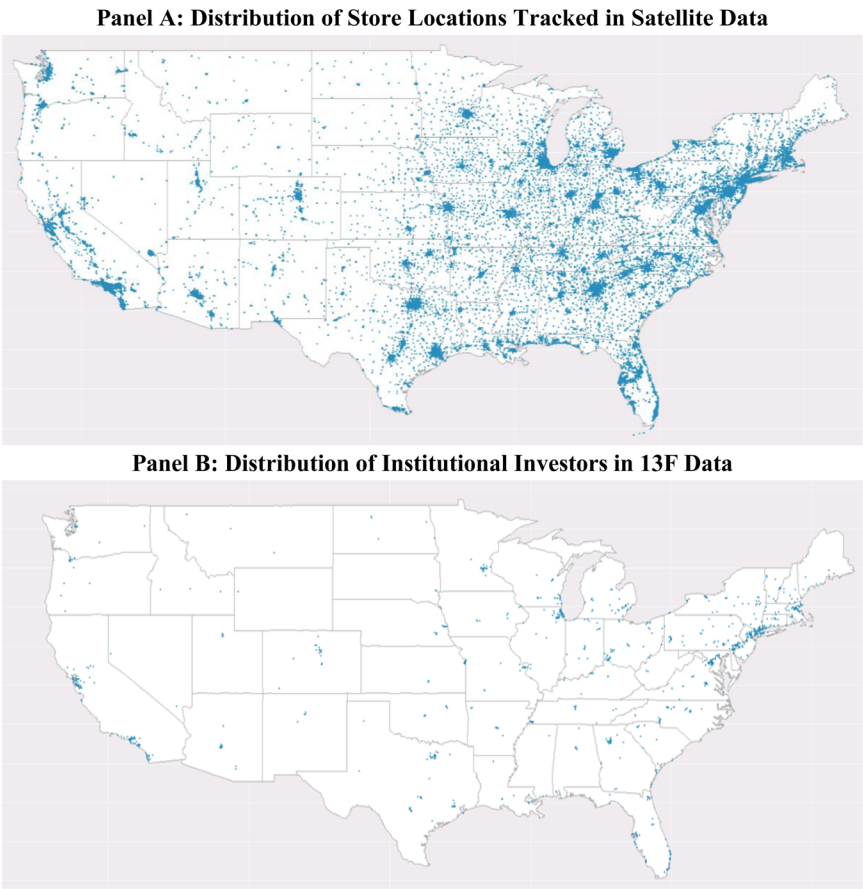


FIG 2.—Distribution of stores and investors across the continental United States. Panel (a): Distribution of store locations tracked in satellite data. Panel (b): Distribution of institutional investors in 13F data. This figure plots locations of the stores (panel a) and investors (panel b) included in our sample across the continental United States.

employees, or be aware of local events or policies that could shift demand. To the extent that institutional investors’ local signals are correlated with local car counts, through their joint relation with local performance, we can use local car counts to examine how investors respond to local store performance. This granular store-level performance proxy is a key element of our design. Even when firms voluntarily disclose disaggregated data, it is not provided in real time or at the store level (Curtis, Lundholm, and Mcvay [2014]). As a result, the car count data have the potential to capture real-time performance information that would not be available through firm disclosures.

To create a timely measure of *new* store-level information (information that investors would be most likely to trade on), we calculate the percentage

change in car counts for a given store compared to the same store in the previous quarter:

$$Car_Chg_{r,s,q} = \frac{(Car_Count_{r,s,q}) - (Car_Count_{r,s,q-1})}{(Car_Count_{r,s,q-1})} \times 100, \quad (1)$$

where r refers to the retailer, s identifies an individual store, and q is the fiscal quarter. Car_Count is the average time-, day-, and weather-adjusted car count for a given store-quarter.¹²

Although car counts are available at the store level, investors must make their stock allocation decisions at the firm level. To reflect this firm-level decision, we aggregate store-level car counts by firm for each investor. Specifically, we calculate the distance-weighted aggregate value of Car_Chg for stores within the firm that are local to the investor (Car_Chg_Local), where local stores receive a higher weight when they are closer to the investor.¹³ Distance weighting decreases noise in our measure by giving lower weight to stores that are relatively farther away, while giving a higher weight to the closest stores, about which local investors would have the greatest amount of information. We also calculate the change in car counts aggregated across *all* stores (Car_Chg_Firm). These two measures reflect the different information signals potentially available to each investor by measuring local performance (Car_Chg_Local) and firm-wide performance (Car_Chg_Firm).

$$Car_Chg_Local_{i,r,q} = \sum_{s=1}^{n_{Local\ i,r,q}} Car_Chg_{r,s,q} \times Weight_{i,r,s,q}, \quad (2)$$

where

$$Weight_{i,r,s,q} = \frac{1/Distance_{i,r,s,q}}{\sum_{j=1}^{n_{r,q}} 1/Distance_{i,r,j,q}}.$$

$$Car_Chg_Firm_{r,q} = \frac{\sum_{s=1}^{n_{r,q}} Car_Count_{r,s,q} - \sum_{s=1}^{n_{r,q}} Car_Count_{r,s,q-1}}{\sum_{s=1}^{n_{r,q}} Car_Count_{r,s,q-1}} \times 100. \quad (3)$$

The subscripts are as above, with the addition of i , which denotes investor. Note that Car_Chg_Firm (Car_Chg_Local) varies at the firm-quarter level (investor–firm-quarter level).

3.3 INVESTOR HOLDING DATA

We use 13F data from Thomson Reuters and WRDS SEC Analytics to measure investor holdings. Form 13F contains a breakdown of the number and value of shares held by institutional investors for each company in which

¹² An average is used because each store is observed an average of 2.83 times within a given quarter.

¹³ As shown in the online appendix, table A1, results are similar if we use unweighted average local car counts. Car_Chg_Local is only defined when the investor is local to at least one store.

they own stock. Thomson Reuters has historically been one of the leading 13F databases. However, after the SEC required XML reporting of 13Fs in 2013, the Thomson data have been subject to a variety of well-documented errors. As a result, we use 13F data from WRDS SEC Analytics for periods after June 2013.^{14,15}

We measure the extent to which investor i 's holdings in firm r changed over quarter q as:

$$Chg_Holdings_{i,r,q} = \frac{(Shares_{i,r,q} - Shares_{i,r,q-1}) \times Price_{r,q}}{\sum_{f=1}^n (Shares_{i,f,q} \times Price_{f,q})} \times 10,000, \quad (4)$$

where $Shares$ is defined as the number of shares held by investor i in firm r in calendar quarter q , and f stands for all of the firms in which investor i has holdings. That is, we measure an investor's change in holdings in a company scaled by their total holdings in that quarter, in basis points.¹⁶

Table 2 and figure 2(b) provide descriptive information about the institutional investors in our sample. The sample includes 2,354 unique investors who each hold 224 firms (21 retailers) on average, or an average of about 14,000 sample stores. These investors hold an average of \$6.5 billion (\$32 million) in total shares (shares in sample retailers). The sample covers 89% (72%) of institutional investors in the 13F universe in terms of dollar holdings (number of investors). Thus, although the *firms* in the sample represent a small fraction of publicly listed firms, the set of *investors* captures the vast majority of U.S. institutional investors. Figure 2(b) shows that institutional investors in the sample are relatively dispersed. Although they are concentrated around large cities, sample institutional investors are present in all states except for Hawaii and Washington, DC.

3.4 DESCRIPTIVE STATISTICS

Table 3 provides descriptive statistics for the variables used in our analyses. Although our tests are conducted mainly at the investor-firm-quarter level, the variables are measured at multiple levels. For example, some variables vary at the firm-quarter level, such as *Sales_Chg* (with a mean of 2.654) and others at the investor-quarter level, such as *Fund_Size* (with a mean of 15.621). We also use investor- and firm-level data, such as *Chg_Holdings* (with a mean of 2.395) and *Car_Chg_Local* (with a mean of 2.354). Because our analyses are conducted at a more aggregate level, some control variables have identical values across observations. We address this by clustering our standard errors (see each table for details).

¹⁴Information on the data errors provided by WRDS is available on the "Research Resources" section of Lorien Stice-Lawrence's Web site, currently found at: sites.google.com/usc.edu/sticelaw/research-resources. These issues are also discussed in Ben-David et al. [2021].

¹⁵Inferences are similar using only Thomson Reuters data.

¹⁶This equation uses the *current* price of firm r and other stocks held by the investor. Because only current values are used, price fluctuations relative to the prior quarter do not mechanically lead to changes in *Chg_Holdings*.

TABLE 2
13F Investor Holdings Data

Panel A: 13F data sample selection			
	Investor–Firm- Quarter Observations	Unique Investors	Unique Firms
(1) Full 13F data set (March 2009 to December 2017)	28,956,311	6,666	9,377
(2) Eliminate observations missing investor location data from 13F (including those where business and mailing address differ) and price data from CRSP	14,619,507	3,256	8,121
(3) After merging with satellite data	404,653	2,497	71
(4) After requiring nonmissing <i>Car_Chg_Local</i> in the current quarter ^a	291,909	2,429	71
(5) Retain pre-period before data were released (up to and including June 2015)	202,453	2,354	71
Panel B: Sample investors versus others			
	13F Investors in Our Sample	13F Investors Not in Our Sample with Available CRSP and Location Data	%
Total number of managers	2,354	902	72%
Total assets under management (\$billions, summing latest per-investor AUM)	15,308	1,841	89%
Average number of unique firms held	224	93	

This table provides information about the 13F investor sample used in the paper. Panel A presents the sample selection process. Panel B compares characteristics of investors in our sample with those of other 13F investors.

^aNot all stores have car counts available in each quarter, and as a result we lose 112,744 investor–firm-quarter observations that lack local store car counts. However, only 68 unique investors were not local to any stores throughout the entire sample.

3.5 VALIDATION OF CAR COUNT MEASURES AS TIMELY PERFORMANCE SIGNALS

In table 4, we test whether our car count measures are valid proxies for firm- and store-level performance. This is important to establish because in our subsequent tests we will examine the extent to which institutional investors respond to local versus nonlocal performance, as captured by the car count data. If our proxies for local and firm-level performance are not correlated with actual performance, it would call into question the interpretation of these tests.

In table 4, panel A, we show that aggregate car counts (*Car_Chg_Firm*) are associated with quarterly changes in sales (column 1), changes in net income (column 3), earnings announcement abnormal return (column 5), and three-month abnormal returns following the earnings announcement (column 7). To ease interpretation, we normalize the independent

TABLE 3
Descriptive Statistics

	N	Mean	Q1	Median	Q3	SD
Firm level						
<i>Car_Chg_Firm</i>	1,218	0.687	−3.629	0.157	4.260	7.097
<i>Avg_Car_Chg_Local</i>	1,218	0.139	−0.978	0.200	0.934	2.025
<i>AbRet_3m</i>	1,218	−0.922	−12.220	−0.875	9.990	18.033
<i>AbRet_3m_Calendar</i>	1,218	−0.199	−12.040	−0.580	11.380	16.938
<i>Analyst_Num</i>	1,218	16.236	8.000	17.000	23.000	8.732
<i>CAR_2d</i>	1,218	0.352	−4.210	0.270	4.930	7.845
<i>Forecast_Vol</i>	1,218	0.325	0.043	0.086	0.206	0.952
<i>Leverage</i>	1,218	1.737	0.584	1.002	1.885	5.701
<i>Log_Assets</i>	1,218	7.544	6.196	7.307	8.619	1.525
<i>Log_Mktcap</i>	1,218	7.640	6.362	7.484	8.881	1.598
<i>MTB</i>	1,218	4.345	1.521	2.495	4.074	19.973
<i>NL_Chg</i>	1,218	−17.693	−59.866	−11.087	37.624	265.033
<i>Ret_Vol</i>	1,218	0.218	0.035	0.080	0.201	0.444
<i>ROA</i>	1,218	1.690	0.675	1.655	2.753	2.249
<i>Sales_Chg</i>	1,218	2.654	−6.004	0.881	9.882	17.621
<i>Store_Num</i>	1,218	5.865	4.820	5.652	6.807	1.147
Investor level						
<i>Aggr_Car_Chg_Local</i>	34,109	0.598	0.045	0.253	0.601	8.055
<i>Fund_Size</i>	34,109	15.621	12.194	13.068	14.427	1.681
<i>Fund_Return</i>	34,109	3.677	0.000	3.526	7.972	6.479
<i>High_Social_Ties</i>	34,109	0.430	0.000	0.000	1.000	0.495
<i>Quasi_Indexer</i>	34,109	0.318	0.000	0.000	1.000	0.466
<i>Performance</i>	29,976	0.473	0.000	0.000	1.000	0.499
<i>Specialist</i>	34,109	0.545	0.000	1.000	1.000	0.498
Store level						
<i>Car_Chg</i>	654,874	12.013	−19.399	0.119	24.803	86.527
<i>Nearby_Car_Chg</i>	654,874	2.557	−1.783	0.000	2.419	94.076
<i>Poor_Signal</i>	419,122	0.305	0.000	0.000	1.000	0.460
Investor–Firm Level						
<i>Chg_Holdings</i>	202,453	2.395	−0.829	0.000	1.355	2.860
<i>Car_Chg_Local</i>	202,453	2.354	−1.127	0.457	3.384	11.471
<i>Poor_Local_Signal</i>	173,115	0.207	0.000	0.000	0.000	0.405
<i>Trade_Local</i>	202,453	0.518	0.000	1.000	1.000	0.500
<i>Fund Weight</i>	202,453	5.579	0.244	3.223	7.244	16.104
<i>Proximate</i>	202,453	1.451	0.000	1.386	2.398	1.306
Investor–store level						
<i>Local</i>	179,374,540	0.019	0.000	0.000	0.000	0.136

This table presents descriptive statistics for all of the variables employed in our analysis. All variables are described in detail in the appendix. All continuous variables are winsorized at the 1% and 99% level. All variables used in regressions are normalized by their respective means and standard deviations reported in this table.

variables in these and all regressions by subtracting their mean and dividing by their standard deviation. The coefficient of 5.150 on *Car_Chg_Firm* in column 1 indicates that a one-standard-deviation increase in firm-level car count changes is associated with a sales increase of 5.2%. This is after controlling for current and past abnormal returns ($AbRet_{3m_q}$, $AbRet_{3m_{q-1}}$) and car count changes in the same quarter last year ($Car_Chg_Firm_{q-4}$) to

TABLE 4
Association of Satellite Car Count Data with Concurrent Firm Performance

Panel A: Analysis using car count data aggregated to the firm level							
	Sales_Chg _q		NI_Chg _q		CAR_2d _q		AbRet_3m _{q+1}
	(1)	(2)	(3)	(4)	(5)	(6)	
<i>Car_Chg_Firm_q</i>	5.150*** (5.65)		23.356** (2.47)		0.393*** (3.06)		0.416*** (2.80)
<i>Avg_Car_Chg_Local_q</i>	0.169 (0.41)	1.442* (1.85)	-1.524 (-0.31)	3.250 (1.36)	0.096 (1.01)	0.245* (1.76)	0.028 (0.54)
<i>Log_Assets_q</i>	-0.819 (-0.21)	-0.913 (-0.21)	-148.391** (-2.09)	-148.816** (-2.04)	0.035 (0.03)	0.027 (0.02)	-8.561*** (-5.75)
<i>Log_Mktcap_q</i>	-0.525 (-0.34)	0.127 (0.08)	64.630* (1.87)	67.588* (1.95)	-1.869*** (-3.80)	-1.820*** (-3.66)	6.405*** (12.27)
<i>AbRet_3m_q</i>	-0.009 (-0.31)	-0.012 (-0.37)	-0.752 (-1.22)	-0.764 (-1.22)	0.000 (0.01)	-0.000 (-0.01)	0.027*** (3.12)
<i>AbRet_3m_{q-1}</i>	0.030 (0.87)	0.056 (1.62)	-0.390 (-0.73)	-0.271 (-0.52)	0.012 (1.59)	0.014* (1.77)	0.063*** (10.45)
<i>MTB_q</i>	0.003 (0.22)	0.006 (0.41)	0.232 (0.44)	0.245 (0.48)	0.004 (0.92)	0.004 (0.99)	-0.003 (-0.87)
<i>Ret_Vol_q</i>	-4.317*** (-2.97)	-4.969*** (-2.95)	38.755 (1.10)	35.800 (1.00)	-0.126 (-0.23)	-0.176 (-0.33)	0.314 (0.87)
<i>Analyst_Num_q</i>	0.051 (0.31)	0.021 (0.13)	1.118 (0.47)	0.985 (0.41)	-0.071 (-1.18)	-0.073 (-1.21)	-0.223*** (-2.98)

(Continued)

TABLE 4—(Continued)

	Sales_Chg _q		NI_Chg _q		CAR_2d _q		AbRet_3m _{q+1}	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Forecast_Vol _q	2,729** (2.57)	3,099** (2.38)	-17,847 (-0.70)	-16,168 (-0.62)	0.542*** (3.59)	0.571*** (3.67)	0.316* (1.69)	0.339* (1.68)
Store_Num _q	2,680 (0.56)	2,059 (0.42)	14,672 (0.25)	11,857 (0.20)	-0.184 (-0.22)	-0.231 (-0.28)	-0.974 (-1.45)	-1.012 (-1.50)
Car_Chg_Firm _{q-4}	-0.251 (-0.85)	-0.245 (-0.84)	-2,006 (-0.59)	-1,977 (-0.59)	0.036 (0.52)	0.037 (0.53)	0.013 (0.34)	0.014 (0.35)
Fixed effects	Quarter, firm	Quarter, firm	Quarter, firm	Quarter, firm	Quarter, firm	Quarter, firm	Quarter, firm	Quarter, firm
Observations	1,218	1,218	1,218	1,218	1,218	1,218	1,218	1,218
Adjusted R ²	0.096	0.045	0.067	0.059	0.055	0.050	0.531	0.525

(Continued)

TABLE 4—Continued

Panel B: Analysis using car count data disaggregated to the store level								
	Sales_Chg _q		NI_Chg _q		CAR_2d _q		AbRet_3m _{q+1}	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Car_Chg _q	0.270 ^{***} (4.93)	0.257 ^{***} (3.57)	1.139 [*] (1.81)	1.152 [*] (1.93)	0.051 ^{***} (2.89)	0.048 ^{**} (2.57)	0.084 ^{***} (5.06)	0.068 ^{***} (5.31)
Nearby_Car_Chg _q	0.012 (0.27)	0.013 (0.23)	−0.001 (−0.43)	−0.392 (−0.43)	−0.025 (−1.12)	0.011 (0.54)	−0.000 (−1.54)	−0.009 (−0.34)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	State-quarter, firm	State-quarter, firm	State-quarter, firm	State-quarter, firm	State-quarter, firm	State-quarter, firm	State-quarter, firm	State-quarter, firm
Observations	654,874	355,161	654,874	355,161	654,874	355,161	654,874	355,161
Adjusted R ²	0.106	0.138	0.125	0.138	0.143	0.173	0.602	0.627

The purpose of this table is to demonstrate the association between car count measures “aggregated to the firm level” with concurrent measures of firm performance for the same quarter. Firm-quarter-level regressions estimate the association between *Car_Chg_Firm* and *Avg_Car_Chg_Local* with firm performance (*Sales_Chg*, *NI_Chg*, *CAR_2d*, *AbRet_3m_{q+1}*), where *Car_Chg_Firm* is the average change in store-level car counts, *Car_Chg*, across all stores in a firm-quarter and *Avg_Car_Chg_Local* is the average value of *Car_Chg_Local* for all investors in our sample with holdings in that firm-quarter. The purpose of this table is to demonstrate the association between car count measures “disaggregated to the store level” with concurrent measures of firm performance for the same quarter. Store-quarter-level regressions estimate the association between *Car_Chg* (store-level changes in car counts) with firm performance (*Sales_Chg*, *NI_Chg*, *CAR_2d*, *AbRet_3m_{q+1}*). Columns 1, 3, 5, and 7 include all values of *Car_Chg* (all those used to calculate *Car_Chg_Firm*) while columns 2, 4, 6, and 8 include only those stores that are local to at least one investor (those used to calculate *Car_Chg_Local*). These store-level specifications also control for general economic conditions of each store by including state-quarter fixed effects and a control for the average performance of all other stores within 50 km (*Nearby_Car_Chg*). We cluster standard errors in this panel by firm-quarter to account for repeated values of firm-quarter control variables across stores within a firm-quarter. Control variables comprise all those included in Panel A. All continuous independent variables are normalized by their standard deviation. Robust *t*-statistics clustered by firm-quarter are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

control for seasonality. Therefore, an individual with real-time car count data could predict performance before it was announced. In addition, panel A provides evidence that *local* car counts are associated with firm outcomes. Because *Car_Chg_Local* is defined at the investor–firm level while performance is measured at the firm-level, we define *Avg_Car_Chg_Local* as the average value of *Car_Chg_Local* in that firm-quarter. The results in columns 2, 6, and 8 indicate that local car counts are significantly related to changes in sales, announcement abnormal returns, and future abnormal returns, consistent with local car counts being useful information signals. The significance level of these results is weaker than for *Car_Chg_Firm*, and *Avg_Car_Chg_Local* is not related to changes in net income (column 4). Further, columns 1, 3, 5, and 7 show that *Avg_Car_Chg_Local* has no incremental explanatory power above and beyond *Car_Chg_Firm*. These results are not surprising and indicate that investors would be better off having information about all of a firm’s stores rather than a subset.

In panel B, we replicate panel A at the store level and show that *Car_Chg*, the change in car counts for an individual store, is significantly positively related to all 4 firm-level outcome variables, even when we restrict *Car_Chg* to the subset of stores that are local to at least one investor (columns 2, 4, 6, and 8). The benefit of conducting these tests at the store level is that we can control for the local economic environment of each individual store using the average value of *Car_Chg* for all other stores located within 50 km (*Nearby_Car_Chg*) and state-quarter fixed effects. The fact that we still find significant results indicates that store-level car counts capture store-specific performance information, not just local economic conditions.¹⁷

In summary, table 4 indicates that car count data capture real-time store-level performance information. In contrast, prior studies, such as Bernile, Kumar, and Sulaeman [2015], have relied on indirect measures of local performance (e.g., based on a portfolio of nearby firms) using reported financial data that would only be available to investors after the quarter end.

4. Empirical Results

4.1 TESTS OF H1 AND H2: INFORMATION ADVANTAGE RELATING TO THE PERFORMANCE OF LOCAL STORES

To test H1, we estimate the following investor- and firm-level regression:

$$\begin{aligned} Chg_Holding_{i,r,q} = & \alpha + \beta_1 Car_Chg_Local_{i,r,q} + \beta_2 Car_Chg_Firm_{i,r,q} \\ & + CONTROLS. \end{aligned} \quad (5)$$

β_1 captures the extent to which investors adjust their holdings in response to the performance of local stores incremental to their response to

¹⁷ Similar to panel A, if we include *Car_Chg_Firm* as an additional regressor, its coefficient is significantly positive while that of *Car_Chg* becomes insignificant, consistent with firm-level signals being less noisy than store-level ones.

overall firm performance (β_2). H1 predicts $\beta_1 > 0$. Both β_1 and β_2 may be insignificant if investors do not have access to performance information in real time or if the signals that they observe are not correlated with car counts.

By examining *changes* in investor holdings and car counts, we control for the past relationship of the investor–firm pair and focus on information obtained during the current quarter. Further, testing the effect of *Car_Chg_Local* incremental to *Car_Chg_Firm* constitutes a within-investor–firm-quarter comparison that captures investors’ responses to local information, incremental to information they have about the firm as a whole.¹⁸ Table 5 estimates this equation and also includes the aggregate car counts of all stores that an investor is local to (*Aggr_Car_Chg_Local*) to control for the local economic environment, last quarter performance (ROA_{q-1}) to control for public performance information, and the change in firm-level car counts four quarters earlier to control for seasonality (*Car_Chg_Firm_{q-4}*). Additionally, we control for other firm (*Log_Mktcap*, *Leverage*, *MTB*, *Store_Num*, *AbRet_3m_Calendar*), investor (*Fund_Size*, *Fund_Return*, *Fund_Weight*), and investor–firm (*Proximate*) attributes. We include no fixed effects, firm and state, firm and state-quarter, and firm-quarter and state-quarter fixed effects, respectively. All firm-quarter-level variables (including *Car_Chg_Firm*) drop out when firm-quarter fixed effects are included. Standard errors are clustered by investor and firm.

Consistent with H1, table 5 shows that the coefficient of *Car_Chg_Local* is positive and significant in all columns. The coefficient for *Car_Chg_Firm* is insignificant, although ROA_{q-1} has a positive and significant coefficient in columns 2 and 3, indicating that investors respond to publicly available firm-level information. The greater ability of *Car_Chg_Local* to explain investor trades relative to *Car_Chg_Firm* is likely not due to the fact that *Car_Chg_Local* is more informative about fundamentals; table 4 shows that *Car_Chg_Firm* is more highly correlated with firm outcomes. These results suggest that investors respond to current information about local stores but do not have access to real-time information about firm-level performance.¹⁹ Further, this local information is store-specific and

¹⁸ *Car_Chg_Firm* is never identical to *Car_Chg_Local*. Focusing on investors with both local and nonlocal stores permits a within investor–firm pair comparison but excludes some *across* investor–firm pair comparisons. Untabulated results are similar if we expand the sample to include investor–firm pairs with only nonlocal stores by setting *Car_Chg_Local* to zero for these pairs.

¹⁹ Another explanation is that investors have access to firm-wide information but fail to respond to it because local information is more salient. In table 10, we show that the response to firm-level car counts increases and the incremental response to local car counts disappears when investors gain access to the satellite data used in this paper. This is inconsistent with investors ignoring nonlocal information because it is less salient.

TABLE 5
Local Information Advantage from Local Performance Information

	Dep Var = <i>Chg_Holdings_q</i>			
	(1)	(2)	(3)	(4)
<i>Car_Chg_Local_q</i>	0.555*** (4.18)	0.472*** (3.78)	0.407*** (3.74)	0.379*** (3.84)
<i>Car_Chg_Firm_q</i>	-0.077 (-0.99)	-0.092 (-0.54)	-0.006 (-0.04)	
<i>Aggr_Car_Chg_Local_q</i>	-0.233 (-1.33)	-0.070 (-0.90)	-0.060 (-0.87)	-0.063 (-0.89)
<i>ROA_{q-1}</i>	-0.223 (-1.49)	0.643*** (3.15)	0.468** (2.45)	
<i>Log_Mktcap_{q-1}</i>	0.243 (1.03)	-6.626*** (-4.93)	-0.637 (-0.82)	
<i>Leverage_{q-1}</i>	-1.352*** (-6.48)	-0.018 (-0.79)	-0.023 (-0.85)	
<i>MTB_q</i>	-0.048* (-1.82)	-0.068 (-0.91)	0.057 (0.94)	
<i>Fund_Size_q</i>	-0.391*** (-2.71)	-1.798*** (-6.72)	-1.650*** (-6.10)	-1.631*** (-6.06)
<i>Fund_Return_q</i>	-1.664*** (-7.32)	-0.587*** (-4.51)	-2.325*** (-6.54)	-2.304*** (-6.57)
<i>Car_Chg_Firm_{q-4}</i>	-0.246** (-2.18)	-0.019 (-0.52)	-0.056 (-0.90)	
<i>Store_Num_q</i>	-0.021 (-0.55)	-3.411*** (-4.54)	0.625 (0.63)	
<i>Proximate_q</i>	-0.202 (-1.00)	-0.113 (-0.69)	-0.125 (-0.84)	-0.161 (-1.11)
<i>AbRet_3m_Calendar_{q-1}</i>	0.149 (0.72)	0.427* (1.96)	-0.083 (-0.41)	
<i>Fund_Weight_q</i>	0.288*** (4.15)	0.284*** (4.10)	0.285*** (4.09)	0.286*** (4.10)
Fixed effects	None	Firm, state	Firm, state-quarter	Firm-quarter, state-quarter
Observations	202,453	202,453	202,453	202,453
Adjusted R ²	0.018	0.024	0.032	0.039

This table presents investor-firm-quarter level regressions that show that investors' holdings are related to changes in the car counts of local firms (*Car_Chg_Local*) even after controlling for their response to changes in overall firm car counts (*Car_Chg_Firm*). Firm-quarter observations are linked with the first value of *Chg_Holdings* reported on or after the fiscal quarter end. All continuous independent variables are normalized by subtracting their mean and dividing by their standard deviation. Robust *t*-statistics are clustered by investor and firm in parentheses.
*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

incremental to information about the local economic environment as captured by *Aggr_Car_Chg_Local* and state-quarter fixed effects.²⁰

²⁰ β_1 of 0.379–0.555 indicates that a one-standard-deviation increase in local car count changes is related to a 0.379–0.555 basis point increase in holdings, or about a 9.4–13.8% increase in holdings in that particular firm, which translates to \$964,338 to \$1,412,158. Overall, the response to local information appears to be material but not astronomical. However, Van Nieuwerburgh and Veldkamp [2009] predict even small differences in initial information

Table 6 further validates the role of local performance information by examining whether the main result in table 5 varies systematically based on factors likely to affect investors' acquisition and use of local information, as predicted in H2. First, we examine how local information advantage is affected by social ties. We obtain the prevalence of social clubs and organizations in each institutional investor's local state from the Northeast Regional Center for Rural Development and use this as a proxy for the strength of investors' social ties with their local community (Hasan et al. [2017]). H2a predicts that local investors in regions with high social ties will have a greater information advantage. The results in columns 1 and 2 show the coefficient on *Car_Chg_Local* is significantly higher in regions with high social ties (significant at the 1% level), suggesting that social ties are an important source of information about local operations.²¹

Next, we examine whether institutional investors with industry expertise are more responsive to local information about firms in their industry (H2b). We define investors who specialize in the retail industry (the industry of all firms in our sample) as those whose proportion of holdings in retail firms is above the sample median (*Specialist* = 1). Columns 3 and 4 of table 6 show that retail industry specialists are significantly more responsive to local information about the firms in our sample than nonspecialists, consistent with industry expertise allowing them to analyze and integrate local information at a lower cost.

Third, we use SEC Form ADV to identify institutional investors who charge performance fees because these investors likely have greater incentives to identify and trade on local information (H2c). Consistent with this argument, columns 5 and 6 show that institutional investors who charge performance fees (*Performance* = 1) are significantly more responsive to local information than those who do not.²²

Finally, we divide institutional investors into quasi-indexers and all others (Bushee [2001]). Because quasi-indexers use a passive investment strategy, H2d predicts that they will respond less, if at all, to local performance information. Table 6, columns 7 and 8, shows that nonindexers respond strongly to local information, while the response by quasi-indexers is statistically insignificant (the difference is significant at the 1% level).

4.2 TESTS OF H3: PROFITABILITY OF TRADES BASED ON LOCAL PERFORMANCE INFORMATION

Although tables 5 and 6 provide evidence that investors trade on local performance information, these results are not sufficient to infer that

endowments can lead to large information asymmetries between local and nonlocal investors, and the results of Dyer [2018] suggest that local information also allows investors to use *public* information more profitably.

²¹ We find similar results if we instead measure the strength of social ties using state migration flows.

²² Of the 2,354 investors in our sample, we were able to link only 2,017 with Form ADV data. As a result, our sample decreases to 166,448 observations for this test.

TABLE 6
Cross-Sectional Tests of Local Information Advantage

	Dep Var = $Chg_Holdings_{it}$											
	High_Social_Ties			Specialist			Performance			Quasi_Indexer		
	Yes	Firm-quarter, State-quarter	Yes	Yes	Firm-quarter, State-quarter	Yes	Yes	Firm-quarter, State-quarter	Yes	Yes	Firm-quarter, State-quarter	Yes
$Car_Chg_Local_{it}$												
	= 1	= 0	= 1	= 0	= 1	= 0	= 1	= 0	= 1	= 0	= 1	= 0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	0.731***	0.167*	0.623***	0.137*	0.524***	0.214*	0.843***	0.043	0.843***	0.043	0.843***	0.043
	(3.70)	(1.72)	(3.85)	(1.75)	(2.92)	(1.74)	(3.86)	(0.60)	(3.86)	(0.60)	(3.86)	(0.60)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter	Firm-quarter, State-quarter
Observations	86,998	115,455	97,980	104,473	81,799	84,649	65,551	136,902	65,551	136,902	65,551	136,902
Adjusted R^2	0.049	0.029	0.045	0.054	0.046	0.041	0.064	0.030	0.064	0.030	0.064	0.030
	$F\text{-test } \beta_1(1) = \beta_1(2): 8.62^{***}$			$F\text{-test } \beta_1(3) = \beta_1(4): 12.77^{***}$			$F\text{-test } \beta_1(5) = \beta_1(6): 3.29^*$			$F\text{-test } \beta_1(7) = \beta_1(8): 8.89^{***}$		

This table presents investor-firm-quarter level regressions estimating the specification in column 4 of table 5 for various subsamples of the data in order to explore local information advantage. Columns 1 and 2 compare the response to local car counts between investors in high and low social tie states. Columns 3 and 4 compare the response to local car counts between investors who specialize in the retail industry with those who do not. Columns 5 and 6 compare the response to local car counts between investors who change performance fees and those who do not. Columns 7 and 8 compare the response to local store information between quasi-indexer and nonindexer investors (Bushee 2001). Each F -test tests the two-sided hypothesis that the coefficients of $Car_Chg_Local_{it}$ in each set of columns are equal. All continuous independent variables are normalized by subtracting their mean and dividing by their standard deviation. Robust t -statistics clustered by investor and firm are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

TABLE 7
Profitability of Trades Based on Local Performance Information

	<i>Trade_Local_q</i> = 1 (a)	<i>Trade_Local_q</i> = 0 (b)	Difference (a) – (b)
	(<i>N</i> = 28,186)	(<i>N</i> = 27,357)	
<i>Avg</i>	1.249**	0.761	0.488***
(<i>Weighted_Abn_Ret_{q+1}</i>)	(2.27)	(1.37)	(3.44)
	Annualized difference in returns:		1.952

This table examines the profitability of trades based on local information. It compares the average abnormal returns of the holdings of investors who trade in the same direction as their local signal *Car_Chg_Local* (*Trade_Local* = 1) with the average abnormal returns of the holdings of investors who trade in the opposite direction of *Car_Chg_Local* (*Trade_Local* = 0). Firm-level abnormal returns (*AbRet_3m_Calendar_{q+1}*) are calculated using four-factor risk-adjusted returns (Carhart [1997]) estimated over the three-month period starting on the 13F filing date (i.e., after the quarter in which investors traded). *Weighted_Abn_Ret* is then the weighted average of *AbRet_3m_Calendar* at the investor-quarter level for all of an investor's investments where *Trade_Local* = 1 and where *Trade_Local* = 0, respectively, where firm-level abnormal returns are weighted by the investor's current holdings in each firm. See the appendix for more details. Robust *t*-statistics clustered by investor- and year-quarter are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

local performance information is a source of local information advantage. If local performance information contributes to local information advantage, then investors should earn higher returns when they trade on their local information than when they trade against it. Table 7 fills this gap by comparing the average *Weighted_Abn_Ret* when investors trade with their local information (*Trade_Local* = 1) with the average for when they trade against their local information (*Trade_Local* = 0). *Trade_Local* is an indicator variable set to 1 if *Chg_Holdings* and *Car_Chg_Local* have the same sign. The average value of *Trade_Local* is 0.52, meaning that investors trade in the same direction as their local information slightly more than half the time. *Weighted_Abn_Ret* is the value-weighted average abnormal returns for a given subset of an investor's holdings:

$$\begin{aligned} & \text{Weighted_Abn_Ret}_{i,q,Subset} \\ &= \sum_{r=1}^{n_{Subset}} (AbRet_{3m_Calendar_{r,q+1}} \times Hold_Weight_{i,r,q}), \end{aligned} \quad (6)$$

where

$$Hold_Weight_{i,r,q} = \frac{Shares_{i,r,q} \times Price_{r,q}}{\sum_{j=1}^{n_{Subset}} Shares_{i,j,q} \times Price_{j,q}},$$

where *AbRet_3m_Calendar_{r,q+1}* is the abnormal returns of firm *r* over the three-month period after the investor's holdings are reported, and *Subset* refers to either the subset of holdings where the investor followed their local information (*Trade_Local* = 1) or the subset where they did not (*Trade_Local* = 0). *Weighted_Abn_Ret* is weighted by the dollar holdings in each investment as of the beginning of the return period so that larger holdings have a greater effect on an investor's weighted return. If the investor places higher weights on holdings with higher returns and lower

weights on holdings with lower returns, then they will generate a higher *Weighted_Abn_Ret*.

Table 7 shows that the difference in the average *Weighted_Abn_Ret* between holdings where *Trade_Local* = 1 and those where *Trade_Local* = 0 is 0.488% and statistically significant. This translates to an annualized difference of about 2%. In other words, on average investors earn an additional 2% in abnormal returns each year when they trade based on their local store information. This amount is in line with the magnitude of prior studies documenting the gains of local information (e.g., Bernile, Kumar, and Sulaeman [2015], Coval and Moskowitz [2001]).²³

4.3 TESTS OF H4: DO INVESTORS RELY TOO HEAVILY ON THEIR LOCAL INFORMATION?

We test H4, which examines whether institutional investors incorporate information about the quality of their local information signals, by identifying a subset of local signals that are of ex ante low quality. Table 8 validates our ex ante measure of signal quality by replicating the store-level analysis of table 4, panel B, which links *Car_Chg* to firm-level outcomes, this time including the interaction of *Car_Chg* with *Poor_Signal*. *Poor_Signal* is an indicator variable set to 1 if the sign of *Car_Chg* did not agree with the sign of abnormal returns after the quarter (*AbRet_3m_{t+1}*) in more than two of the previous four quarters. This measure is intended to capture whether the car count signal would have been useful in guiding past trading decisions. The coefficient of *Car_Chg* x *Poor_Signal* is negative for all four outcomes and significant for three of the four, indicating that *Poor_Signal* accurately identifies poor-quality signals. Further, the sum of the coefficients of *Car_Chg* and *Car_Chg* x *Poor_Signal* is insignificantly different from 0 for three of the four outcomes, including both return outcomes (columns 3 and 4), meaning that individual car count signals no longer provide profitable trading information when *Poor_Signal* = 1.

Table 9 aggregates signal quality to the investor–firm level, *Poor_Local_Signal*, by comparing the sign of abnormal returns (*AbRet_3m_{t+1}*) with the sign of investors' local information (*Car_Chg_Local*).²⁴ Panel A examines whether investors discount poor-quality local information (H4). We re-estimate column 4 of table 5 with the addition of *Car_Chg_Local* x *Poor_Local_Signal* and expect the coefficient to be significantly negative if investors discount poor-quality local signals. In contrast, the coefficient is insignificantly different from zero. This suggests that on average

²³ Untabulated tests also show that stronger local signals lead to larger returns for those who trade on them.

²⁴ *Poor_Signal* and *Poor_Local_Signal* require at least two quarters of data but flag signals if they disagree with returns in more than two quarters. This design choice maximizes sample size but works against our ability to find results.

TABLE 8
Validating Measure of Poor Store-Level Performance Signals

	<i>Sales_Chg_q</i> (1)	<i>NI_Chg_q</i> (2)	<i>CAR_2d_q</i> (3)	<i>AbRet_3m_{q+1}</i> (4)
<i>Car_Chg_q</i> × <i>Poor_Signal_q</i>	−0.213*** (−3.20)	−2.260 (−1.46)	−0.083* (−1.93)	−0.113** (−2.12)
<i>Car_Chg_q</i>	0.549*** (4.80)	2.596* (1.92)	0.078** (2.03)	0.097*** (3.62)
<i>Poor_Signal_q</i>	0.498 (0.66)	−9.562 (−1.12)	−0.002 (−0.01)	0.072 (0.70)
Controls	Yes	Yes	Yes	Yes
Fixed effects	State-quarter, firm	State-quarter, firm	State-quarter, firm	State-quarter, firm
Observations	419,122	419,122	419,122	419,122
Adjusted <i>R</i> ²	0.111	0.137	0.157	0.643

This table examines whether our ex ante measure of poor signal quality, *Poor_Signal*, accurately identifies store-level car count changes (*Car_Chg*) that have a poor ability to predict current firm-level performance outcomes (*Sales_Chg*, *NI_Chg*, *CAR_2d*, *AbRet_3m_{t+1}*). *Poor_Signal* is an indicator variable set to 1 if the sign of *Car_Chg* disagreed with the sign of *AbRet_3m_{t+1}* in more than two of the prior four quarters. Control variables comprise all those included in table 4. All continuous independent variables are normalized by subtracting their mean and dividing by their standard deviation. Robust *t*-statistics clustered by firm-quarter are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

institutional investors do not significantly adjust the extent to which they rely on local information when it is of poor quality.²⁵

To explore why sophisticated institutional investors appear to disregard local signal quality, we examine variation in the net benefits that institutional investors expect to gain from using their local signals. Table 6 showed that institutional investors who specialize in the retail industry are more responsive to local information about our sample firms, perhaps because they are better able to interpret this information. If industry expertise also decreases the cost of evaluating local signal quality, then these investors may be less likely to trade on poor-quality local information. To test this, column 1 of table 9, panel B, replicates table 9, panel A, this time including the triple interaction *Car_Chg_Local* × *Poor_Local_Signal* × *Specialist*, and shows that the coefficient is significantly negative. This suggests that industry specialists significantly reduce their response to ex ante poor-quality local information on average.²⁶

²⁵ To ensure this insignificant result is not due to a lack of power, Table A4 in the online appendix examines whether overreliance on poor-quality local information leads institutional investors to make less profitable trading decisions, because investors may still be better off trading on poor-quality local information as long as it provides incremental information. Consistent with institutional investors relying too heavily on local information on average, Table A4 the online appendix finds that investors who trade on poor-quality local information have significantly lower returns than if they had ignored their local information.

²⁶ Table 6 also showed variation in investors' responses to local information based on whether they were quasi-indexers or located in regions with strong social ties. However, these attributes are unlikely to be tied to how investors differentially respond to high- and low-quality

TABLE 9
The Extent to Which Institutional Investors Rely on Poor-Quality Local Information

Panel A: Effect of signal quality		Dep Var = $Chg_Holdings_q$
$Car_Chg_Local_q \times Poor_Local_Signal_q$		-0.033 (-0.50)
$Car_Chg_Local_q$		0.324*** (3.04)
$Poor_Local_Signal_q$		-0.120 (-0.31)
Controls		Yes
Fixed effects		Firm-quarter, state-quarter
Observations		173,568
Adjusted R^2		0.031
Panel B: Variation in the response of institutional investors		
	(1)	(2)
	Dep Var = $Chg_Holdings_q$	
$Car_Chg_Local_q \times Poor_Local_Signal_q \times Specialist_q$	-0.146** (-2.60)	
$Car_Chg_Local_q \times Poor_Local_Signal_q \times Performance_q$		-0.139** (-2.51)
$Car_Chg_Local_q \times Specialist_q$	0.298* (1.79)	
$Car_Chg_Local_q \times Performance_q$		0.466*** (2.82)
$Car_Chg_Local_q \times Poor_Local_Signal_q$	0.072** (2.36)	0.065 (1.40)
$Poor_Local_Signal_q$	0.050 (0.13)	0.102 (0.27)
$Car_Chg_Local_q$	0.179** (2.28)	0.139* (1.93)
$Specialist_q$	1.372*** (5.51)	
$Performance_q$		0.979*** (2.97)
Controls	Yes	Yes
Fixed effects	Firm-quarter, state-quarter	Firm-quarter, state-quarter
Observations	173,568	142,663
Adjusted R^2	0.031	0.031

This table examines whether investors decrease the weight they place on local information (Car_Chg_Local) if their local information is of ex ante poor quality ($Poor_Local_Signal = 1$). Panel A re-estimates the specification from column 4 of table 5 with the addition of $Car_Chg_Local \times Poor_Local_Signal$. Panel B further includes $Car_Chg_Local \times Poor_Local_Signal \times Split$, where $Split$ refers to either $Specialist$ (institutional investors who are specialists in the retail industry) or $Performance$ (institutional investors who charge performance fees) to see whether some investors are more sensitive to local signal quality. Control variables comprise all those included in table 5. All continuous independent variables are normalized by subtracting their mean and dividing by their standard deviation. Robust t -statistics clustered by investor and firm are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

In addition, table 6 showed that institutional investors who charge performance fees respond more to local information than those who do not. These investors stand to benefit personally from high performing investments and therefore have greater incentives to acquire and use high-quality local information. As a result, we expect they will be less likely to trade based on low-quality local signals. Again column 2 of table 9, panel B, replicates table 9, panel A, this time including the triple interaction *Car_Chg_Local* x *Poor_Local_Signal* x *Performance*. The coefficient is significantly negative, indicating that on average investors who charge performance fees respond less to local signals that are of ex ante poor quality.

Overall, the results of table 9 indicate that institutional investors do not mitigate their response to poor-quality local information on average, but do rely less upon poor-quality local signals when the net benefits of determining signal quality are higher. These results suggest that processing costs may be one reason why institutional investors sometimes rely upon poor-quality local information (Blankespoor et al. [2019]). However, we cannot rule out the possibility that some of this behavior is driven by behavioral biases (Tversky and Kahneman [1971, 1974]).

4.4 EFFECT OF SATELLITE DATA AVAILABILITY ON LOCAL INFORMATION ADVANTAGE

Investors may have used the Orbital Insight data to overcome some of their nonlocal information disadvantage after they became available for purchase. We examine whether they did so using data from the post-commercial availability period that was previously excluded from our tests. Because of confidentiality agreements, Orbital Insight was not able to share a list of their institutional investor clients. However, they were able to provide the identities of sell-side research firms who purchased access to the data. Goldman Sachs gained access to the data on March 27, 2017, and JP Morgan gained access on March 20, 2017.²⁷ We link these firms with individual institutional investors by identifying investors for which one of these two firms was the primary broker.²⁸ If brokers incorporate car count information into their analyst reports, models, or recommendations, or if they convey this information to their clients directly, then institutional investor clients should be able to mitigate the part of their nonlocal information disadvantage that is captured by car count information after their primary broker gains access to Orbital Insight.

local signals. First, indexer investors do not appear to use local information (whether of high or low quality) at all, and there is no reason to expect that the strength of social ties in an investor's local community would be tied to their ability or incentive to determine local signal quality.

²⁷ RBC gained access on June 30, 2018, but this falls outside of our sample period that ends in December 2017.

²⁸ We identify an institutional investor's primary broker from the TASS service provider data set from 2016. As a result, the sample in this test includes only hedge funds that reported to TASS.

TABLE 10
Effect of Satellite Data Availability on Local Information Advantage

	Dep Var = $Chg_Holdings_q$	
	(1) Pre- and Post-Period	(2) Post-Period Only
$Car_Chg_Firm_q \times Post_q$	0.502*** (4.08)	
$Car_Chg_Local_q \times Post_q$	-1.916** (-2.52)	
$Car_Chg_Firm_q$	0.104 (0.33)	0.387*** (4.42)
$Car_Chg_Local_q$	1.012** (2.10)	-1.004 (-1.10)
Controls	Yes	Yes
Fixed effects	Firm, state-quarter	Firm, state-quarter
Observations	8,292	595
Adjusted R^2	0.117	0.231

The purpose of this table is to examine the effect that investors' access to the Orbital Insight data had on the local information advantage captured by car counts. Column 1 re-estimates the specification from column 3 of table 5 with the addition of the interactions $Car_Chg_Firm \times Post$ and $Car_Chg_Local \times Post$, where $Post$ identifies quarters after an investor's primary broker had purchased access to Orbital Insight; post-period data were excluded from all previous tests. Column 2 re-estimates the specification from column 3 of table 5, but focuses only on the period when $Post = 1$. The sample in both columns is restricted to only those investors whose primary broker gained access to Orbital Insight during our sample period. Control variables comprise all those included in table 5. All continuous independent variables are normalized by subtracting their mean and dividing by their standard deviation. Robust t -statistics clustered by investor and firm are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table 10, column 1, shows that institutional investors become more responsive to firm-level car count changes after they gain access to Orbital Insight ($Car_Chg_Firm \times Post > 0$). The main effect on Car_Chg_Local shows that these investors respond to local car counts in the pre-period. Similarly, column 2 shows that they respond *only* to firm-level car counts in the post-period, with no incremental response to local car counts. These results suggest that nonlocal investors use car count data to mitigate their information disadvantage and stop relying on local signals when firm-level signals are available. Of course, this does not mean there is no longer any difference between the information sets of local and nonlocal investors, only that the signal conveyed by car counts is equally accessible to local and nonlocal investors with access to Orbital Insight.

These preliminary results add to the growing literature that documents that big data can be used to mitigate information asymmetries and inform stakeholders (Kang [2020], Zhu [2019]), potentially leveling the information playing field. However, our results show only that nonlocal information disadvantage decreased for institutional investors who gained access to the data; meanwhile, this could further exacerbate information differences with retail investors or smaller institutional investors who cannot afford to purchase the data (Katona et al. [2019]).

4.5 RULING OUT ALTERNATIVE CHANNELS AND EXPLANATIONS

We briefly mention results of tests that explore alternative explanations for our findings. These tests, as well as others, are tabulated and described in the online appendix. First, we investigate whether our results are driven by reverse causality. We examine whether changes in local institutional holdings lead to greater subsequent local car counts, for example, as a result of heightened monitoring, but find no evidence that such a relation exists (table A2 in the online appendix). We investigate whether local news media is the main channel for our results (Engelberg and Parsons [2011], Miller and Shanthikumar [2015]) by examining whether our results are mitigated around negative shocks to local news media (local newspaper closures and mass journalist layoffs) that have been shown to negatively impact firms' information environment (Kim et al. [2019]). However, we find no change in investors' response to local car counts after these events, suggesting local news media are not the main source of local information in our setting (table A3 in the online appendix).

5. Conclusion

We use satellite image data on the number of cars in retail store parking lots to study the local information advantage of institutional investors. We establish car counts as a timely proxy for store performance and demonstrate that institutional investors' trades are incrementally associated with the performance of local stores, consistent with local investors having greater information about nearby operations. The response to local information is strongest: in regions with tightly knit social networks; for retail industry specialists; for institutional investors who charge performance fees; and for nonindexer institutional investors. In addition, trades based on local information are profitable. Our results suggest that information about local store performance is one source of local information advantage. However, we also find evidence that investors rely too heavily on their local information and continue to trade on local signals that are not significant predictors of future returns. This overreliance on poor local information is reduced for institutional investors who charge performance fees (have higher benefits of making profitable trades) and for those who are industry specialists (face lower costs to evaluate information about firms in their specialty industry). This finding is consistent with institutional investors expending efforts to evaluate the quality of their local signals when there is a net benefit of doing so. Finally, we demonstrate that when the satellite data used in our study became commercially available, those who gained access no longer exhibited signs of nonlocal information disadvantage. Although prior research provides indirect evidence of a local information advantage (Coval and Moskowitz [2001]), we measure one source of this advantage, local performance information, and provide evidence on how investors use this information once they have acquired it.

Our study is subject to important caveats. First, we cannot observe investors’ information sets and must infer their information by observing their trades. Second, our results may have limited generalizability because we examine only institutional investors of retail firms, and examine only the local information that is correlated with car counts. Local information advantage may be different for nonretail firms, for other investors, or for other types of local information, for example production- or acquisition-related information that is unlikely to be tied to car counts. We note that while our sample of firms is small, our set of investors represents a substantial portion (approximately three fourths) of the institutional investor universe. Although we acknowledge these caveats, we believe that our study is able to provide important insights with respect to local information.

APPENDIX A: VARIABLE DEFINITIONS

Variable	Definition	Source
<i>Aggr_Car_Chg_Local</i>	The aggregate car change signal for <i>all</i> stores that are local to an investor, including those the investor does not have holdings in, within a given quarter. Calculated as the distance-weighted aggregate car count change of all stores that are located within 50 km in that quarter: $\sum_{s=1}^{n_{ALLLocal\ i,q}} Car_Chg_{i,s,q} \times Weight_{i,s,q}, \text{ where}$ $Weight_{i,s,q} = \frac{1/Distance_{i,s,q}}{\sum_{j=1}^{n_q} 1/Distance_{i,j,q}}.$	Orbital Insight
<i>Analyst_Num</i>	Total number of analysts following a firm.	IBES
<i>AbRet_3m</i>	Three-month cumulative abnormal return (CAR) <i>prior to the firm’s</i> earnings announcement date for the current quarter (%), based on the Carhart [1997] four-factor model.	CRSP, Eventus
<i>AbRet_3m_Calendar</i>	Three-month cumulative abnormal return (CAR) leading up to the 13F report date (%), based on the Carhart [1997] four-factor model.	CRSP, Eventus
<i>Avg_Car_Chg_Local</i>	<i>Car_Chg_Local</i> averaged across all investor–firm pairs in a given firm-quarter.	Orbital Insight
<i>CAR_2d</i>	Two-day cumulative abnormal return (CAR) based on Fama–French four-factor model after and including the firm’s earnings announcement date (%).	CRSP, Eventus
<i>Car_Chg</i>	Percentage change in store-level car counts in the current quarter relative to the prior quarter: $\frac{(Car_Count_{r,s,q}) - (Car_Count_{r,s,q-1})}{(Car_Count_{r,s,q-1})} \times 100.$	Orbital Insight

(Continues)

Variable	Definition	Source
<i>Car_Chg_Firm</i>	Aggregate change in car counts for all of a retailer's stores within a firm-quarter: $\frac{\sum_{s=1}^{n_{r,q}} Car_Count_{r,s,q} - \sum_{s=1}^{n_{r,q-1}} Car_Count_{r,s,q-1}}{\sum_{s=1}^{n_{r,q}} Car_Count_{r,s,q-1}} \times 100.$	Orbital Insight
<i>Car_Chg_Local</i>	Distance-weighted aggregate <i>Car_Chg</i> for all stores that are local to the investor within a firm-quarter: $\sum_{s=1}^{n_{Local\ i,r,q}} Car_Chg_{r,s,q} \times Weight_{i,r,s,q}, \text{ where}$ $Weight_{i,r,s,q} = \frac{1/Distance_{i,r,s,q}}{\sum_{j=1}^{n_{r,q}} 1/Distance_{i,r,j,q}}.$	Orbital Insight, 13F
<i>Car_Count</i>	The average time-, day-, and weather-adjusted car count for a store in a given quarter.	Orbital Insight
<i>Chg_Holdings</i>	The change in an investor's holding in a firm compared to the prior quarter, scaled by the size of the investor's total portfolio, measured in basis points. Firm-quarter observations are linked with the first value of <i>Chg_Holdings</i> reported as of or after the fiscal quarter end. $\frac{(Shares_{i,r,q} - Shares_{i,r,q-1}) \times Price_{r,q}}{\sum_{j=1}^n (Shares_{i,f,q} \times Price_{f,q})} \times 10,000.$	13F data for Thomson Reuters and WRDS SEC Analytics, CRSP
<i>Forecast_Vol</i>	The standard deviation of the EPS forecasts scaled by the stock price at the prior quarter-end (%).	IBES, CRSP
<i>Fund_Return</i>	Quarterly weighted average return for the investments of an investor.	13F
<i>Fund_Size</i>	The natural logarithm of the total investment amounts of an investor.	13F
<i>Fund_Weight</i>	Weight of the current firm in the investor's portfolio, calculated as the dollar amount of holdings in firm <i>r</i> divided by the total dollar value of the investor's portfolio: $\frac{(Shares_{i,r,q}) \times Price_{r,q}}{\sum_{j=1}^n (Shares_{i,f,q} \times Price_{f,q})} \times 10,000.$	13F, CRSP
<i>High_Social_Ties</i>	A measure of social ties using social clubs. Defined as an indicator variable set to 1 if the total number of social clubs divided by 1,000 people in the population is above the sample median. Values are linearly interpolated to generate values for missing years. Social clubs include: religious, civic, business, political, professional, and labor organizations, and bowling and fitness centers, public golf courses, and sports clubs. See more discussion of these social organizations in HASAN et al. [2017].	Northeast Regional Center for Rural Development, 2009 and 2014 data
<i>Leverage</i>	The ratio of total liabilities (<i>ltq</i>) to the total equity (<i>atq-ltq</i>) of a firm as of the end of the quarter.	Compustat
<i>Local</i>	An indicator variable equal to 1 if an investor is within 50 km of an individual store, 0 otherwise.	13F, Orbital Insight
<i>Log_Assets</i>	The natural logarithm of total assets (<i>atq</i>).	Compustat
<i>Log_Mktcap</i>	The natural logarithm of total market value (<i>mkvaltq</i>).	Compustat
<i>MTB</i>	The ratio of market value (<i>mkvaltq</i>) to total equity (<i>teq</i>).	Compustat

(Continues)

Variable	Definition	Source
<i>Nearby_Car_Chg</i>	The average car change signal for all stores that are located in the immediate area surrounding a given <i>store</i> , including those that belong to different firms, within a given quarter. Calculated as the distance-weighted aggregate car count change of all stores that are located within 50 km in that quarter: $\sum_{x=1}^{n_{ALLLocal\ s,q}} Car_Chg_{x,s,q} \times Weight_{x,s,q}$, where $Weight_{x,s,q} = \frac{1/Distance_{x,s,q}}{\sum_{y=1}^{n_q} 1/Distance_{y,s,q}}$.	Orbital Insight
<i>NI_Chg</i>	Quarterly changes in net income (niq) scaled by the net income (niq) in the prior quarter (%).	Compustat
<i>Performance</i>	An indicator variable equal to 1 if the investor charges a performance fee based on the capital gains of the client's assets.	SEC Form ADV
<i>Poor_Signal</i>	An indicator variable set to 1 if the sign of <i>Car_Chg_q</i> is not equal to the sign of <i>AbRet_3m_{q+1}</i> (abnormal returns in the three months immediately after the end of the quarter) in more than two of the previous four quarters. This measure requires that data for at least two of the prior four quarters be available.	Orbital Insight, CRSP, Eventus
<i>Poor_Local_Signal</i>	An indicator variable set to 1 if the sign of <i>Car_Chg_Local_q</i> is not equal to the sign of <i>AbRet_3m_{q+1}</i> (abnormal returns in the three months after the end of the quarter) in more than two of the previous four quarters. This measure requires that data for at least two of the prior four quarters be available.	Orbital Insight, CRSP, Eventus
<i>Proximate</i>	$\ln(1+N)$ where <i>N</i> is the number of the firm's stores that an investor is <i>Local</i> to in the current quarter.	13F, Orbital Insight
<i>Quasi_Indexer</i>	Indicator variable equal to 1 if the investor was categorized as a quasi-indexer according to Bushee [2001].	Brian Bushee's Web site
<i>Ret_Vol</i>	Standard deviation of daily stock price return during the one year before the earnings announcement date, scaled by the stock price at the prior quarter-end.	IBES, CRSP
<i>ROA</i>	The ratio of net income (niq) to total assets (atq).	Compustat
<i>Sales_Chg</i>	Quarterly changes in sales (saleq) scaled by the sales in the prior quarter (%).	Compustat
<i>Specialist</i>	An indicator variable equal to 1 if the institutional investor is a specialist in the retail industry, defined as those whose proportion of total holdings in the retail industry is above the sample median.	13F, Compustat, CRSP
<i>Store_Num</i>	$\ln(1+N)$, where <i>N</i> is the number of stores in a given firm for which car count data are available in a given quarter.	Orbital Insight

(Continues)

Variable	Definition	Source
<i>Trade_Local</i>	An indicator variable set to 1 if the investor changes their holdings in the same direction as the change in their local car counts (i.e., if <i>Chg_Holdings</i> and <i>Car_Chg_Local</i> have the same sign), and zero otherwise.	Orbital Insight, 13F, CRSP
<i>Weighted_Abn_Ret</i>	Investor-quarter level measure of the weighted average abnormal returns of a subset of the investor's holdings. The subset (e.g., <i>Trade_Local</i> = 1, <i>Poor_Local_Signal</i> = 1, etc.) is specified in the table in which <i>Weighted_Abnormal>Returns</i> are used. The returns are calculated over the three-month period following the 13F report date. The weighted average is calculated as: $\sum_{r=1}^{n_{\text{Subset}}} (AbRet_3m_Calendar_{r,q+1} \times Hold_Weight_{i,r,q}),$ where $Hold_Weight_{i,r,q} = \frac{Shares_{i,r,q} \times Price_{r,q}}{\sum_{j=1}^{n_{\text{Subset}}} Shares_{i,j,q} \times Price_{j,q}}.$	13F data for Thomson Reuters and WRDS SEC Analytics, CRSP, Eventus

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